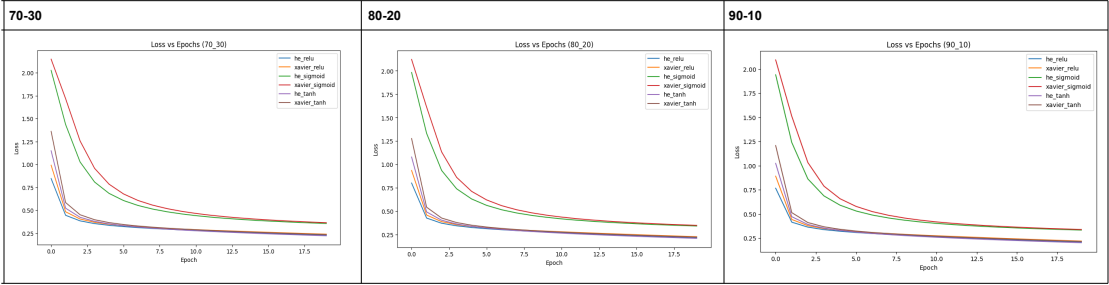


70-30	80-20	90-10
==== 70_30 RESULTS ==== he_relu Acc: 0.9306 xavier_relu Acc: 0.9280 he_sigmoid Acc: 0.9014 xavier_sigmoid Acc: 0.8996 he_tanh Acc: 0.9347 xavier_tanh Acc: 0.9316	==== 80_20 RESULTS ==== he_relu Acc: 0.9364 xavier_relu Acc: 0.9326 he_sigmoid Acc: 0.9033 xavier_sigmoid Acc: 0.9028 he_tanh Acc: 0.9377 xavier_tanh Acc: 0.9346	==== 90_10 RESULTS ==== he_relu Acc: 0.9398 xavier_relu Acc: 0.9368 he_sigmoid Acc: 0.9077 xavier_sigmoid Acc: 0.9053 he_tanh Acc: 0.9420 xavier_tanh Acc: 0.9387



Confusion Matrix	HE and Xavier Initialization with RELU Activation	HE and Xavier Initialization with SIGMOID Activation	HE and Xavier Initialization with TANH Activation
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70-30	70_30 - he_relu [[1722 1 8 3 6 26 19 4 16 0] [0 1947 11 6 3 7 2 3 11 4] [5 19 1603 15 20 12 26 26 25 8] [6 11 36 1605 1 89 7 24 42 25] [1 4 9 1 1642 0 18 4 4 43] [16 7 7 22 11 1540 15 5 19 11] [6 5 11 0 14 25 1714 1 10 1] [5 17 34 2 20 4 0 1813 5 37] [6 31 15 25 12 44 23 3 1554 17] [12 7 6 13 49 14 0 43 8 1611]] 70_30 - xavier_relu [[1722 1 6 3 6 28 16 6 17 0] [0 1942 11 6 3 9 2 2 15 4] [5 19 1600 14 20 14 27 26 26 8] [6 10 38 1597 1 94 8 23 43 26] [1 5 11 2 1636 0 18 3 3 47] [16 9 9 23 10 1533 17 6 20 10] [5 5 11 0 13 26 1715 2 10 0] [5 18 34 1 21 3 0 1813 4 38] [7 35 15 24 12 51 23 3 1543 17] [12 6 7 13 53 15 0 44 10 1603]]	70_30 - he_sigmoid [[1711 0 11 4 5 30 18 4 22 0] [0 1925 13 8 3 14 1 5 23 2] [9 31 1532 19 25 10 46 37 38 12] [9 11 52 1558 1 104 13 26 43 29] [1 6 15 0 1599 4 21 6 7 67] [20 24 12 46 13 1452 28 11 36 11] [11 6 17 0 22 36 1684 3 8 0] [4 39 31 3 24 10 0 1769 6 51] [6 47 24 55 15 74 20 7 1457 25] [16 8 10 21 76 18 1 63 11 1539]] 70_30 - xavier_sigmoid [[1712 0 11 5 5 29 17 4 22 0] [0 1926 12 9 3 13 1 5 23 2] [10 32 1520 19 24 8 50 37 39 12] [10 11 50 1552 0 110 12 28 45 26] [1 8 15 0 1592 5 21 6 7 71] [21 26 13 48 14 1443 33 10 33 12] [16 6 20 0 22 36 1678 1 8 0] [4 41 31 3 22 12 0 1771 4 49] [4 50 25 54 15 79 20 5 1452 26] [17 8 11 21 73 19 1 65 10 1538]]	70_30 - he_tanh [[1732 1 7 3 6 21 14 4 16 1] [0 1943 13 7 3 9 1 2 13 3] [5 17 1612 15 18 11 24 26 24 7] [5 8 38 1633 1 67 7 24 42 21] [1 4 10 0 1648 0 15 4 3 41] [13 11 8 30 10 1536 14 6 16 9] [4 5 7 0 14 21 1724 2 10 0] [4 17 26 3 19 4 0 1827 4 33] [6 26 14 27 13 47 18 6 1558 15] [11 7 5 15 54 11 0 41 7 1612]] 70_30 - xavier_tanh [[1730 1 7 3 6 23 13 4 17 1] [0 1940 13 7 3 9 1 2 16 3] [5 17 1606 17 18 14 23 26 25 8] [5 8 38 1617 1 80 8 23 43 23] [1 4 9 2 1646 0 14 4 4 42] [13 10 8 30 9 1530 14 7 21 11] [4 5 6 0 14 23 1725 1 9 0] [4 18 26 2 17 5 0 1824 4 37] [7 28 14 29 13 55 20 6 1542 16] [13 6 7 15 51 11 0 44 7 1609]]	
	80-20	80_20 - he_relu [[1136 0 5 1 3 9 8 2 9 2] [0 1288 8 6 1 3 1 2 11 2] [1 11 1086 12 10 6 12 14 16 6] [3 5 21 1116 1 25 6 7 15 20] [1 3 6 2 1118 0 8 4 1 33] [14 5 7 27 12 1006 9 5 13 6] [7 2 9 0 9 17 1125 1 7 0] [4 10 23 3 10 1 0 1221 4 23] [6 17 7 33 11 21 13 3 1041 8] [8 5 5 11 25 6 0 30 4 1100]] 80_20 - xavier_relu [[1132 0 4 1 3 9 8 4 12 2] [0 1284 8 5 1 4 1 2 14 3] [1 11 1083 11 12 8 12 14 15 7] [3 5 22 1110 1 27 6 10 18 17] [1 3 6 2 1113 0 9 4 2 36] [14 5 6 32 11 1001 9 5 15 6] [7 2 10 0 9 18 1122 2 7 0] [4 11 23 5 10 1 0 1219 3 23] [6 18 8 33 10 23 15 3 1036 8] [9 5 5 12 31 6 0 31 4 1091]]	80_20 - he_sigmoid [[1119 0 6 3 4 16 9 4 14 0] [0 1271 8 10 1 8 1 3 18 2] [3 20 1034 13 19 5 27 22 24 7] [8 6 33 1057 1 48 8 11 27 20] [1 5 9 1 1079 1 13 5 3 59] [19 14 14 49 12 939 12 5 33 7] [13 3 8 0 11 15 1117 4 6 0] [5 30 23 4 14 4 0 1187 5 27] [6 27 15 45 11 38 13 5 987 13] [10 6 8 16 38 12 1 44 9 1050]] 80_20 - xavier_sigmoid [[1121 0 7 2 4 16 8 4 13 0] [0 1270 7 11 1 8 1 3 19 2] [3 18 1038 12 18 4 28 21 24 8] [8 6 34 1056 0 48 7 11 28 21] [1 5 9 0 1083 0 13 5 3 57] [22 14 14 49 13 933 14 5 31 9] [14 4 12 1 12 13 1114 1 6 0] [4 30 23 4 15 4 0 1188 4 27] [5 28 15 45 12 39 14 4 985 13] [11 6 8 16 39 12 1 46 9 1046]]	80_20 - he_tanh [[1136 0 3 1 4 10 7 4 9 1] [0 1287 10 5 1 3 1 1 12 2] [1 11 1086 12 11 5 14 15 15 4] [3 4 25 1112 1 24 6 12 18 14] [1 2 5 3 1125 0 8 5 2 25] [15 6 5 36 8 1005 7 5 12 5] [4 2 5 0 9 16 1132 2 7 0] [3 14 17 3 9 2 0 1229 2 20] [7 16 10 28 10 23 12 5 1044 5] [8 6 4 12 29 5 0 29 4 1097]] 80_20 - xavier_tanh [[1134 0 3 1 4 10 9 4 9 1] [0 1284 10 5 1 4 1 1 13 3] [1 11 1079 12 12 7 15 13 18 6] [4 4 26 1104 1 28 6 11 20 15] [1 3 4 3 1126 0 7 4 2 26] [15 6 6 35 8 1002 7 6 13 6] [5 2 7 0 9 16 1130 2 6 0] [3 14 17 4 10 2 0 1227 3 19] [6 16 10 29 10 27 12 5 1038 7] [9 5 5 13 30 6 0 30 5 1091]]

90-10

```
90_10 - he_relu
[[602 0 2 1 1 4 5 1 7 1]
 [ 0 632 4 5 0 0 0 2 10 1]
 [ 1 2 535 7 5 3 2 6 9 2]
 [ 0 2 13 548 0 12 1 2 6 5]
 [ 1 2 4 1 547 0 3 2 1 19]
 [ 6 3 3 16 9 501 4 2 5 2]
 [ 7 0 4 0 5 7 553 1 3 0]
 [ 1 9 10 2 4 1 0 592 4 10]
 [ 3 7 8 11 3 4 7 1 540 1]
 [ 4 2 2 6 13 3 0 8 5 589]]
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```
90_10 - xavier_relu
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 [ 0 629 5 3 0 2 0 2 12 1]
 [ 1 2 535 7 6 3 2 5 9 2]
 [ 0 2 16 543 0 13 1 2 8 4]
 [ 1 2 4 1 547 0 4 1 1 19]
 [ 6 4 3 16 9 500 4 2 5 2]
 [ 7 0 4 0 5 7 553 1 3 0]
 [ 1 9 10 3 3 1 0 592 4 10]
 [ 4 7 7 13 3 7 5 0 538 1]
 [ 4 2 2 6 17 3 0 9 6 583]]
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90_10 - he_sigmoid
[[597 0 2 1 1 10 7 0 6 0]
 [ 0 628 4 7 0 4 0 2 8 1]
 [ 2 4 513 6 10 2 10 7 13 5]
 [ 1 2 17 519 0 23 1 5 12 9]
 [ 1 4 5 1 530 0 7 2 0 30]
 [ 10 9 8 20 6 469 8 2 18 1]
 [ 8 1 5 0 8 5 549 1 3 0]
 [ 3 16 11 3 6 3 0 575 5 11]
 [ 4 16 9 21 4 16 6 1 503 5]
 [ 4 2 3 8 20 6 0 19 7 563]]
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```
90_10 - xavier_sigmoid
[[596 0 2 1 1 11 7 0 6 0]
 [ 0 630 4 6 0 3 0 2 8 1]
 [ 2 4 512 6 10 2 11 7 13 5]
 [ 1 2 18 519 0 23 0 5 13 8]
 [ 1 4 5 1 526 0 7 2 0 34]
 [ 13 11 8 21 7 462 9 2 16 2]
 [ 9 1 6 0 9 4 548 0 3 0]
 [ 3 17 10 3 6 3 0 575 5 11]
 [ 4 16 9 22 4 17 6 1 502 4]
 [ 4 2 3 8 21 7 0 18 7 562]]
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```
90_10 - he_tanh
[[601 0 2 1 1 6 6 1 6 0]
 [ 0 635 5 4 0 1 0 1 7 1]
 [ 1 2 540 6 5 3 0 4 9 2]
 [ 0 2 17 543 0 11 1 3 6 6]
 [ 1 1 3 2 551 0 3 2 0 17]
 [ 5 4 4 16 5 504 5 3 5 0]
 [ 5 0 4 0 6 6 555 0 4 0]
 [ 1 9 8 2 3 1 0 597 4 8]
 [ 4 7 6 13 3 8 6 1 536 1]
 [ 4 2 1 6 14 3 0 7 5 590]]
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```
90_10 - xavier_tanh
[[600 0 1 1 1 6 8 1 6 0]
 [ 0 633 5 3 0 2 0 1 9 1]
 [ 1 2 535 7 7 3 0 5 8 4]
 [ 2 2 18 537 0 11 1 3 9 6]
 [ 1 2 4 2 551 0 2 1 0 17]
 [ 6 4 4 16 5 501 3 3 7 2]
 [ 5 0 4 0 6 6 556 0 3 0]
 [ 1 10 7 3 3 1 0 597 3 8]
 [ 4 7 6 13 3 10 6 1 533 2]
 [ 4 2 1 6 16 3 0 7 4 589]]
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